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PI Name:	Phillips, James O Ph.D.		
Project Title:	Gravitational Influence on Cerebellar Control of Gaze Movement and Adaptation		
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PI Email:	jop@u.washington.edu	**Fax:**	FY 206 616-1828
PI Organization Type:	UNIVERSITY	**Phone:**	206 543-0265
Organization Name:	University of Washington		
PI Address 1:	CD176F CHDD Building		
PI Address 2:	VM Bloedel Hearing Research Center		
PI Web Page:			
City:	Seattle	**State:**	WA
Zip Code:	98195	**Congressional District:**	7
Comments:			
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Contact Monitor:		**Contact Phone:**	
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COI Name (Institution):	Fuchs, Albert F. (University of Washington) Robinson, Farrel R. (University of Washington)		
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Task Description:	Micro-gravity alters the normal relationships between coordinated multi-axial body movements and sensory feedback. As subjects undergo transitions between 1G to near 0G, they experience inappropriate eye and head movement and disorientation when shifting gaze. These changes pose a substantial challenge to mission objectives. One effort to address such challenges has been to develop context dependent preadaptation training protocols to train astronauts to quickly adapt based on inertial cues. Unfortunately, while this approach holds much promise, little is known about the reference frame for such adaptation or the underlying physiological mechanisms responsible for these adaptive changes. This black box approach limits progress toward the development of appropriate countermeasures. Our approach is to develop a laboratory analog of the altered gravitational/inertial environments, and then use identical psychophysical tests to evaluate behavioral performance of human and animal subjects in these environments to define both the neural		

mechanism and the crucial factors determining adaptation.
Our research will consist of 3 components. First, we will describe the characteristics of context dependent human adaptation and its transfer to other forms of adaptation, seeking to localize behaviorally the site of adaptation in the CNS. Second, we will perform identical experiments on rhesus monkeys while recording the neural activity of cells within the caudal fastigial nucleus, which is thought to be important for saccadic adaptation. Third, we will simulate changes in the vestibular environment by changing the activity of the vestibular end organ in monkeys. We will study context dependent adaptation and the neural discharge associated with adaptation in this altered vestibular environment.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

We began our project on 2/27/04 with the receipt of our award. We received preliminary approval for our work with human subjects on 2/24/04, and we received full approval on 9/17/2004.
We have made significant progress on each aspect of our proposed research project in the first year of approval and we are on target to complete the entire project on schedule. Our progress has resulted in 2 posters and a platform presentation at international meetings, and we have 3 manuscripts in preparation based on our findings to date. Each group working on this project has contributed significant findings. In addition, in related activity, we responded to the loss of the primary supplier of human scleral coil recording devices with the development of a new technology for eye and head movement recording. Furthermore, we are collaborating with the group of Dr. Millard Reschke at Johnson Space Center, Houston, on the interpretation of human saccadic eye movement data collected before, during, and immediately following spaceflight. Our progress for the first year is documented below.

Human Experiment. (Fuchs, Ling, Hopp, and Phillips) – Our preliminary results suggest that eye and head movements are adapted together during head unrestrained gaze adaptation in humans. Furthermore, the adaptation of human saccadic eye movements in the head restrained laboratory condition transfers to the head unrestrained condition. This result has important operational significance because it suggests that a human preflight saccade adaptation countermeasure will transfer to the context in which natural head unrestrained gaze movements are being performed. Such transfer is critical to any useful countermeasure, since astronauts will not be performing saccadic eye movements under laboratory conditions during 0G to 1 G transitions, but rather will be performing natural head unrestrained gaze shifts.

Human Experiment. (Fuchs and Hopp) – Preliminary results suggest that cognitive volitional saccade adaptation transfers across conditions. For example, transfer occurs between memory guided and delayed saccades. The previous work suggested that human saccadic eye movement adaptation transfers between reflexive saccades, such as targeting saccades and express saccades. There is poor transfer between reflexive saccade adaptation and cognitive saccades. This provides both a benefit and a challenge for operational implementation of a preflight adaptation countermeasure. It provides evidence that there is transfer between adaptation of specific saccade types and other saccade types that were not specifically adapted in the laboratory but which would presumably be utilized during spaceflight. It also suggests, however, that human saccade adaptation does not transfer as widely across saccade types as does non-human primate saccade adaptation. Therefore, multiple preflight adaptations might be required to serve as an effective countermeasure for saccades of different types. The precise specificity remains to be determined.

Animal Experiment. (Phillips, Fuchs, Ling). Elimination of canal afferent input modifies the reflex behavior and discharge of brainstem neurons during head unrestrained gaze shifts in rhesus monkeys. We trained 2 rhesus monkeys to perform head unrestrained gaze shifts and then lesioned the vestibular end organ with the use of bilateral canal plugging. This resulted in a change in the coordination of eye and head movements during head unrestrained gaze shifts. We observed that animals made accurate gaze shifts following canal plugging, but the relationship of saccadic eye movement end and gaze movement end, and the amplitude and duration of the eye and head movement components was dramatically altered. This data suggests that vestibular signals are important for the normal control of gaze movement, but that new strategies emerge, involving adaptation of saccade amplitude, to compensate for the loss of vestibular input and maintain accurate gaze movements after canal plugging. Furthermore, we recorded cells in the brainstem burst generator before and following bilateral canal plugging. The results suggest that vestibular signals converge at both the premotor neuron and motoneuron level to control gaze movements in normal monkeys, and that the burst generator loses the capacity to drive an accurate gaze movement to completion following loss of vestibular input. Other mechanisms, downstream of the pontine burst generator, a putative target of caudal fastigial nucleus inputs, are required to restore accurate gaze movement following vestibular loss. These data are critical for assessing the importance of vestibular signals during gaze shifts with normal and abnormal vestibular contributions. Our data suggests that disruption of vestibular inputs does not prevent the successful adaptation of saccades and the emergence of new control strategies for accurate gaze.

Animal Experiment. (Fuchs and Ling) We have recorded a subset of neurons in the caudal fastigial nucleus (CFN) of 2 rhesus monkeys that display discharge characteristics consistent with a head movement signal during active head unrestrained gaze shifts. This discharge is similar to the head related signal of prolonged burst neurons recorded in the pontine brainstem during head unrestrained gaze shifts, in that it dramatically outlasts the saccade related discharge of saccade neurons in the CFN. Such a neuron could carry the vestibular signals requisite for the control of coordinated gaze movement during head unrestrained gaze shifts. Likewise, these cells are possible targets for the vestibular inputs that may be required to establish the context for context dependent adaptation. We will target this population in future recording experiments during both head unrestrained gaze shifts and tilt context dependent adaptation in the next two years.

Animal Experiment. (Noto, Robinson and Phillips) In rhesus monkeys, saccades made during tilt orientations in pitch and roll are dysmetric, producing dissociated vertical deviation of the eye and, in some monkeys, changes in saccade velocity and amplitude. We recorded the saccades of 3 rhesus monkeys during 45 deg static tilt in pitch and roll and observed the resulting targeting saccades. The saccades were observed to be dissymmetric in all animals, with a vertical deviation of the ipsilateral eye upward and the contralateral eye downward during static roll tilt downward. This result suggests that a vertical diplopia exists during roll tilt in rhesus monkeys which is uncompensated by normal short-term vestibulo-ocular mechanisms. This could pose a significant challenge to mission objectives if similar dissociations are present in humans under similar conditions, and if the diplopia compromises vision under some visual stimulus conditions, i.e., reduced lighting or required fixation of weakly illuminated avionics during disorienting 1G to 0G

transitions.

Animal experiment. (Phillips, Noto and Robinson) We determined, in rhesus monkeys, that the adaptation field during tilt is in craniotopic coordinates and is largely unaffected by tilt orientation. Adaptation of saccade gain is restricted to a limited range of vectors relative to the head, which are directionally maintained during saccade testing in other orientations. We tested the saccades of rhesus monkeys to 10-degree target steps in 8 cardinal directions, and then adapted rightward saccades so that they had reduced gain. We then retested the saccades in 8 directions during roll tilt to the right or left, showing a constant adaptation field relative to the head. If these results can be extended to humans, an important question in year two, the limited adaptation field with respect to the head will suggest an important limitation to preflight adaptation paradigms as a countermeasure. Since the adaptation is restricted to specific vectors with respect to the head, the range of vector directions must be extended during preflight training to allow for adequate representation of visual space during natural movements.

Animal experiment (Phillips, Noto and Robinson). Rhesus monkeys display transfer of adaptation across tilt adaptations, and initial tilt orientation does affect to degree of transfer between orientations. In these experiments, we produced a gain decrease adaptation in one direction along the horizontal meridian with respect to the head at a specific tilt orientation (pitched forward, pitched back, roll right, roll left or upright) and then changed the tilt orientation to another orientation to observe transfer of adaptation. Adaptation transferred to each tilt orientation, with the best transfer occurring when the initial adaptation was in the normal, head upright orientation. Also, the rate constants for adaptation were faster in the head upright condition. If these results can be extended to humans, an important question in year two, this suggests that adaptation training will transfer to a range of tilt orientations or inertial contexts.

Related activity. (Kaneko, Phillips) In year one we learned that the major supplier of scleral search coils for human eye movement recording was suspending production of the devices. We responded by purchasing sufficient supplies of the devices to perform our experiments for the full 3 years of our funding. In addition, however, we began development with a local firm of a replacement printed circuit device that could serve as a cheap alternative to the existing eye movement transducers. Development of this technology is proceeding in association with our sustained research activity, and we are seeking separate funding via an STTR mechanism to complete development and begin production of the device.

Related activity. (Reschke, Phillips) Our ground based research compliments flight research begun many years ago at Johnson Space Center by Millard Reschke and colleagues. Dr. Reschke recorded pre-flight, flight and post-flight saccadic eye movement data in astronauts during several space missions. We have begun a collaboration to reexamine some of this very important flight data to better understand the nature of gaze and eye movement changes related to adaptation to novel inertial environments. Specifically, we are interested in the metrics of saccades and gaze shifts, and in the stability of post saccadic gaze. It is critical that we understand nature of the changes in gaze movement control during the transition to and from 1 G to near 0 G to optimize strategies for applying proposed countermeasures.

Bibliography Type:	Description: (Last Updated: 05/13/2009)
Abstracts for Journals and Proceedings	Phillips, J.O. "Effects of Bilateral Canal Plugging on The Discharge Of Brainstem Neurons during Eye and Head Gaze Shifts: Where Is the Vestibular Signal in Gaze Control?" Barany Society XXIII Meeting Journal of Vestibular Research, 14:276, P181 , Jun-2004
Abstracts for Journals and Proceedings	Phillips, J.O., Noto, C, Robinson, F.R. "The influence of tilt on normal and adapted saccades in monkey " Society for Neuroscience 2004 Annual Meeting SFN Abstracts, p99010 , Oct-2004
Patents	UWTTR# 7137D in process Jan-2005 Kaneko, C.R.S., Phillips, J.O. "Printed Circuit Search Coil for Eye and Head Movement Recording in Humans and Animals"
Presentation	Phillips, J.O. "Vestibular contributions to gaze shifts: lesion studies." Symposium S14, Eye-head-hand coordination Barany Society XXIII Meeting, Jul-2004